# Model Alpha M14 Class Balance Research Brief

HELIX FORGE LAB / p04-full-008556 / RUN R184 / 2026-07-13

| ALPHA F1 | PRECISION | RECALL | BETA F1 |
|---|---|---|---|
| 0.873 | 0.881 | 0.865 | 0.846 |

## Decision context

This report records a bounded class balance evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

## Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

## Lineage

Upstream evidence: p04-full-000148 (research/programs/model-alpha/experiments/configs/2026-07-13/model-alpha-m14/seed-stability/tables/model-alpha-m14-seed-stability-s

| Slice | Records | Precision | Recall | F1 |
|---|---|---|---|---|
| slice-01 | 198 | 0.865 | 0.853 | 0.859 |
| slice-02 | 216 | 0.882 | 0.866 | 0.874 |
| slice-03 | 196 | 0.864 | 0.879 | 0.871 |
| slice-04 | 252 | 0.881 | 0.855 | 0.868 |
| slice-05 | 203 | 0.863 | 0.868 | 0.865 |